
Hit Rate Is Not Value: When Cache Metrics Mislead in Tiered LLM KV-Cache Persistence*

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Abstract

1 We set out to beat LRU at evicting persisted LLM KV caches with a learned
2 policy, and failed: across three synthetic workloads, two served architectures,
3 and a replay of the Azure LLM inference trace, LRU wins cache hit rate every
4 time. The interesting result is what that verdict conceals. Measured in GPU time
5 actually saved, the same LRU delivers the *least* value of any policy we test on
6 enterprise traffic, where a value-density policy conceding 21 hit-rate points saves
7 $1.99\times$ as much compute, and still $1.15\times$ once we correct a provisioning error the
8 same analysis exposes. Hit rate assumes every hit is worth the same; KV caching
9 violates that twice, since a hit below the break-even context length N^* is worth
10 nothing and above it savings grow superlinearly. Both effects are closed-form in
11 N^* , and the largest lever they expose is provisioning, not policy: deleting a tier
12 that cannot repay its restore is worth $3.6\times$ at identical capacity. A ground-truth
13 ablation isolates the failure: two clairvoyant policies that both cache perfectly,
14 100.0% against 100.0%, still deliver significantly different value.

15 1 Introduction

16 Serving a large language model proceeds in two phases: *prefill* processes the prompt in parallel and
17 populates the KV cache, and *decode* generates one token at a time against it. Prefill costs $O(N^2)$
18 attention FLOPs while the cache grows only linearly in N , so the cache makes multi-turn boundaries
19 cheap. VRAM has no room for idle sessions, though: an 80 GB A100 fits only ~ 30 full-context
20 Llama-2-7B sessions after weights, against thousands of users whose think time spans seconds to
21 days. Serving engines therefore evict idle caches [6], and a returning user re-prefills everything.
22 Tiered persistence demotes those caches down a hierarchy (VRAM $\rightarrow$ NVMe $\rightarrow$ object storage),
23 paying $O(N)$ bytes moved instead of $O(N^2)$ FLOPs recomputed [4, 12, 10]. That relocates the
24 eviction problem rather than eliminating it, since every tier is finite.

25 That decision looks like fertile ground for learning. Sessions differ by orders of magnitude in size,
26 their recompute cost grows quadratically where their footprint grows linearly, and whether a user
27 returns looks predictable from history. LRU sees none of this, so we hypothesized that a learned
28 policy estimating resumption probability would beat it on hit rate. *The data falsified this hypothesis*,
29 and diagnosing why became the contribution: LRU wins hit rate on every constrained workload, and
30 no feature engineering closed the gap. But the metric was the problem: measured by GPU time saved,
31 that same LRU delivers the *least* value of any policy on enterprise traffic, and policies losing to it by
32 11–21 hit-rate points save up to twice as much. Hit rate assumes every hit is worth the same, and
33 KV caching breaks that twice: a hit below the break-even length N^* is worth *nothing*, and above it
34 savings grow superlinearly (§4).

35 **Related work.** Systems that persist, disaggregate, or offload KV state across a hierarchy [6, 4, 12, 13]
36 build the persistence mechanism; we target the policy layer above it. Prefix sharing [5, 10, 16] and

*System, committed run records, and the scripts regenerating every number, figure, and table here: <https://github.com/aditya-lawankar/kv-cache-tier-persistence>

token-level compression [17, 9, 7, 8] differ in granularity. Cost-aware eviction is old: GreedyDual-Size [3] and its learned successors [14, 1] already evict by value per byte, and our V2 is GD-Size without aging. What differs is a *sign*: in web and CDN caching every hit is weakly positive, so admission is an optimization, whereas a sub- N^* hit here is net *negative*, which puts the threshold in provisioning and admission rather than in an eviction score.

2 Setup

What a tier holds, and why there is more than one. The unit of management is a whole conversation: one session’s KV tensors form a single object of tens of MB to a few GB. When a session goes idle and VRAM is needed, a serving engine would discard it, forcing a full re-prefill on return; instead we serialize it one tier down and read it back later. The hierarchy exists because capacity and restore cost pull against each other: VRAM is fast and scarce, NVMe restores at 10 ms + 2 GB/s for far less per byte, object storage cheaper still at 100 ms + 500 MB/s. Demotion is triggered by capacity, not by a timer: when a tier is full, its eviction policy picks a victim, which moves down a level or leaves the hierarchy entirely from the last one. Each tier therefore has its *own* restore cost, which is what makes the metric fail: a hit from a slow tier can cost more to restore than the prefill it avoids, and hit rate counts it anyway.

Policies. Six policies replay identical traces, each evicting the entry minimizing its own score. For entry i , let v_i be the GPU time its cached tokens would save, s_i its bytes, $\mathbb{E}[\Delta t_i]$ its estimated time to next access, and f_i, r_i, n_i its access count, decayed idle time and normalized token count. The scores are: *LRU*, last access time; a *recency-frequency heuristic*, $0.4f_i + 0.4r_i + 0.2n_i$; *Logistic V1*, $\hat{P}(\text{resume} \leq 1\text{h})$ from a nine-feature model (test AUC 0.70); *Value Density V2*, v_i/s_i ; *V2+AC*, the same but admitting only entries above a value threshold; and *Space-Time V3*, $v_i/(s_i \mathbb{E}[\Delta t_i])$. V3 divides by expected lifetime so an entry is charged for the space-time it occupies, after LHD [1]. They differ sharply in what they consume: LRU needs only access order, the heuristic adds counts and sizes, V1 a trained model over nine features, V2/V3 the cost model pricing v_i . That widening dependence is what §5 shows makes them fragile.

Workloads and clock. Synthetic profiles use Poisson arrivals and log-normal token counts: enterprise (200/hr, $\mu=2048$, $\sigma=0.5$) and power user (80/hr, $\mu=8192$, $\sigma=1.2$), clamped at 8,192 tokens over a 6-hour window at 500 MB (50/150/300 MB across three tiers); a third casual profile never constrains the cache. Each user carries a persistent persona (return propensity, verbosity, a diurnal window); without one, per-user features are i.i.d. noise and no predictor beats chance. Traces replay on a *virtual clock*: wall-clock replay collapses session age, idle time, and hour-of-day to near zero, invalidating any policy that reasons about time while leaving order-based ones untouched.

Real traces. We also replay the Azure LLM inference trace [11, 15]: 27.3M production requests over one week, whose arrival process and context sizes (median 928, p95 4,943 tokens) transfer verbatim into ten 6-hour windows used as paired replicates. It cannot provide session identity, and linkage is not reconstructable: a greedy context-growth matcher links 87.8% of requests, and an indistinguishable 87.8% under a permutation control that destroys all true chains. We therefore impose return behavior with the same persona machinery, so differences from synthetic come from the arrival process and sizes, which are real. The learned policies run zero-shot.

Value accounting. Each hit is credited the prefill it avoids for the tokens *actually cached*, minus the modeled restore from its tier, both priced by Eq. 1; a hit whose restore exceeds its avoided prefill scores zero. We report GPU-seconds saved per day under both TinyLlama-1.1B (22 KB/token) and Llama-2-70B (320 KB/token, GQA) economics. The GPU-second is the scale-free quantity: dollars are a linear rescaling that depends on a reader’s negotiated price, and at our arrival rates the dollar figures would say more about the size of the simulated deployment than about the policies.² The cost model is parameterized by the served architecture rather than fitted constants, and validated against hardware: over a $12\times$ context sweep on a T4 its predicted prefill tracks measurement within $1.15\text{--}1.66\times$, and its break-even lengths match an independently coded analytic model exactly (App. A). Every cell runs 10 seeds with all policies on the identical per-seed trace, so comparisons are *paired*: per-seed deltas against LRU with 95% t-CIs.

²A sub- N^* hit is credited zero rather than the negative value it actually delivers. Crediting negatives is arguably more correct and would widen every gap we report, since LRU earns by far the most such hits; we floor at zero as the conservative choice.

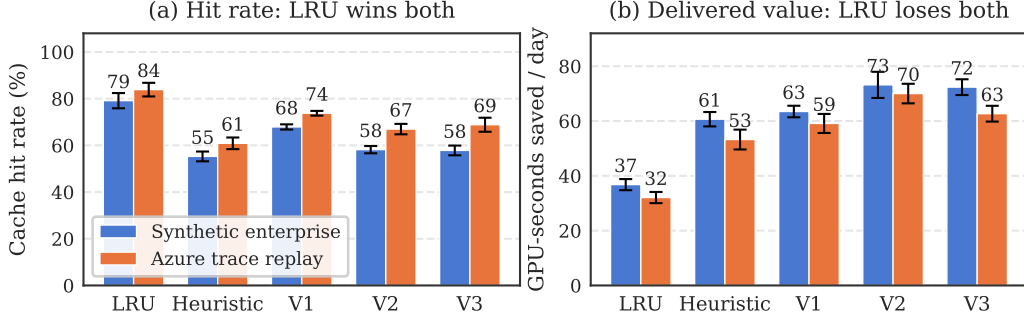


Figure 1: The same runs, two metrics: LRU wins hit rate on both workloads and delivers the least value on both. 10 paired seeds (synthetic) or ten 6-hour windows (Azure), 95% t-CIs, 500 MB.

3 Hit Rate and Value Reverse

LRU wins the metric we set out to beat, taking the highest hit rate on every constrained workload: 79.1% [75.9, 82.4] on enterprise against Logistic V1’s 67.9% [66.8, 69.0], and 96.0% against 83.9% on power user.

That verdict is narrower than it sounds. On enterprise traffic (Figure 1) LRU simultaneously delivers the least value of any policy: V2 saves $1.99\times$ as much GPU time (+36.4 GPU-s/day [+30.4, +42.4]), V3 $1.97\times$, V1 $1.73\times$, and even the untrained heuristic $1.65\times$, every delta significant. Both replicate on real traffic, and the second strengthens: on the Azure replay LRU again wins hit rate (83.9%, by 16.9 points over V2) and again delivers the least value, V2 saving $2.18\times$. The inversion is *larger* on real traffic because real sizes are more skewed, so more mass falls below N^* (§4).

4 Mechanism: A Threshold and a Superlinearity

Restoring a persisted cache moves $O(N)$ bytes, while recomputing its prefill costs $O(N)$ weight FLOPs plus $O(N^2)$ attention:

$$t_{\text{restore}}(N) = \ell_{\text{tier}} + \frac{2LH_{kv}dbN}{B_{\text{tier}}}, \quad t_{\text{prefill}}(N) = \frac{2PN + 4LH_qdN^2}{F\eta}, \quad (1)$$

for L layers, H_{kv} KV heads (GQA shrinks the cache, not the scores, so query heads H_q appear in prefill), head dimension d , b bytes/element, tier latency ℓ , bandwidth B , P parameters, peak F FLOP/s, and utilization η . Restore is linear and prefill superlinear, so they cross at a break-even length N^* , set by KV bytes per token over prefill FLOPs per token. That ratio spans four orders of magnitude: on an A100 at $\eta=0.45$, N^* is 1,561 tokens from NVMe (10 ms, 2 GB/s) and 25,901 from object storage (100 ms, 500 MB/s) for TinyLlama-1.1B; 43,559 and 254,203 for Llama-2-7B; and 12 and 289 for Llama-2-70B, whose GQA shrinks bytes moved while parameters inflate FLOPs avoided. The 7B figure is not a slip: a 512 KB/token MHA cache is so large against its prefill FLOPs that NVMe offload never repays at realistic contexts. That is why systems shipping KV state compress it [8] or use DRAM and RDMA, not a 2 GB/s disk [12]. Our $\eta=0.45$ is generous to prefill; higher MFU pushes N^* out, not in. The threshold is objective-agnostic too: the same inequality governs time-to-first-token.

Hit rate expresses neither effect. Under TinyLlama economics every cold-tier hit in our token range is worth exactly zero, and LRU, blind to size and tier alike, earns 778 of its 1,206 enterprise hits there; above N^* , meanwhile, savings grow superlinearly, so one 8k-token session can outweigh a dozen 2k-token ones.

Table 1 separates them two ways. Repricing at Llama-2-70B, where every hit clears break-even, collapses the enterprise gap to $1.07\text{--}1.08\times$ and hands power-user traffic back to LRU; the threshold thus accounts for roughly $1.85\times$ and the magnitude effect for the residual $\sim 1.08\times$. Deleting the cold tier instead, at fixed architecture and 500 MB, estimates that residual independently at $1.13\text{--}1.15\times$: V1 and V3 each deliver $1.15\times$ more than LRU while conceding 11 hit-rate points. Two manipulations

122 agreeing within two points isolate the same residual superlinearity. Admission control, by contrast,
 123 never helps: V2+AC concedes 4.8 GPU-s/day to plain V2 on enterprise $[-5.3, -4.3]$ and 6.1 on the
 124 Azure replay $[-6.7, -5.4]$, because under sustained load the cache runs near-full and refusing entries
 125 at the door only costs cardinality (App. B).

126 The same table carries this paper’s largest effect, and it is not a policy effect. *Deleting the cold tier*
 127 *raises LRU from 36.8 to 131.0 GPU-s/day, a factor of 3.6, at identical total capacity*; two-tier LRU
 128 beats the best three-tier policy we built by $1.8\times$. The same bytes provisioned differently beat the
 129 same bytes managed better, so the $1.99\times$ carries a condition: it holds for a deployment our own
 130 model advises against building, and the portable claim is the $1.15\times$ surviving correct provisioning,
 131 found only by pricing hits rather than counting them.

132 **Ground truth: identical perfect hit rates, different value.** Perhaps the learned policies lose because
 133 their predictor is weak (AUC 0.70), not because the objective is wrong. We replay the same traces
 134 with *oracle* policies reading the actual future, legitimate because replay is open-loop: Bélády [2],
 135 plus oracle V1 and V3 on ground-truth returns. All three cache perfectly at 100.0%, so every miss in
 136 §3 is a prediction failure, not a capacity necessity, and LRU trails them $5.70\times$ in value against 20.9
 137 hit-rate points.

138 That setup yields the cleanest evidence here: every other source of difference is gone. Hit rate cannot
 139 distinguish Bélády from Oracle-V3: both read 100.0%, a paired difference of exactly zero on every
 140 seed. Their value differs significantly nonetheless, by $+2.9$ GPU-s/day $[+2.0, +3.8]$ on enterprise,
 141 because once nothing need ever be dropped the only decision left is *which tier* holds each entry. The
 142 effect is small (1.4%) but real, and hit rate reports it as exactly zero.

143 5 Discussion and Conclusion

144 **Two bugs, both flattering, both ours.** We corrected our own evaluation twice, in opposite directions:
 145 a wall-clock replay bug flattered the learned policy, an architecture-free cost model flattered LRU
 146 (App. C). The asymmetry is structural. A policy consuming only access order is insensitive to broken
 147 time, features, *and* value models alike; every richer policy is exposed to all three. Harness bugs
 148 therefore flatter whichever side depends on the broken component rather than perturbing rankings at
 149 random, so LRU comparisons controlling neither replay semantics nor cost-model fidelity deserve
 150 re-examination *both* ways.

151 **A decision rule, in order of payoff.** (1) Compute N^* per tier from Eq. 1: published geometry and
 152 storage constants, no measurement. (2) Do not build tiers whose N^* exceeds the workload’s context
 153 lengths, the largest lever we measured at $3.6\times$. (3) Refuse admission below the surviving tiers’ N^* ,
 154 the step our data supports least directly, since density-thresholded admission never helped us and its
 155 payoff arrives as step (2). (4) Only then choose a policy, and never tune on hit rate before establishing
 156 the regime.

157 **Limitations.** Return behavior and value are both modeled rather than observed; Apps. A and D
 158 bound each. No policy we built wins *both* metrics anywhere.

159 **Conclusion.** Where a workload straddles N^* , hit rate ranks KV-cache policies backwards: it picks
 160 the worst one available and calls it best. Beating that metric does not require beating LRU.

Table 1: Two controls decomposing the effect (enterprise, paired seeds; heuristic omitted, see §3).
Middle: repricing at Llama-2-70B moves N^* to 12 tokens, switching the threshold off, and the best
margin over LRU falls from $1.99\times$ to $1.08\times$. **Right:** deleting the cold tier at fixed 500 MB leaves
 $1.15\times$.

Policy	TinyLlama, 3-tier		Llama-2-70B, 3-tier		TinyLlama, 2-tier	
	Hit%	GPU-s/d	Hit%	GPU-s/d	Hit%	GPU-s/d
LRU	79.1	36.8	79.1	8,173	79.4	131.0
Logistic V1	67.9	63.5	67.9	8,806	68.0	150.7
ValDensity V2	58.2	73.2	67.1	8,722	58.5	147.4
SpaceTime V3	57.8	72.4	63.2	8,783	58.1	150.0

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A Cost-model validation

Every value figure in this paper is produced by a model, so the model is the thing most worth attacking. An earlier version of it deserved the attack: it priced prefill with two hardcoded constants and no model geometry, placing the linear/quadratic crossover near 1,300 tokens where real architectures cross between 12 and 254k, and predicting 57,747 ms for a 598-token TinyLlama prefill that a T4 measures at ~ 37 ms. The model used here (Eq. 1) is parameterized by published model geometry and storage constants instead, and we check it two ways.

Against hardware. Table 2 sweeps context length on Llama-3.2-1B (16 layers, GQA 8/32, 32 KB/token) on a T4 and compares predicted prefill against measured cold TTFT. Predictions track measurement within $1.15\text{--}1.66\times$ over a $12\times$ range of context, and the residual is one-sided: measurement is consistently slower than prediction, implying an achieved MFU near 0.22 against the 0.30 assumed. The functional form is what matters for our argument, and it holds: over that range cold prefill grows $14.8\times$ while the warm path grows $1.8\times$.

Against an independent implementation. The break-even lengths quoted in §4 are computed by benchmarks/breakeven_analysis.py from the same expressions written separately, and agree exactly with the simulator’s cost model at all three architectures and both tiers. This catches transcription errors, not modeling ones.

Table 2: Predicted versus measured prefill, Llama-3.2-1B on a T4, five trials per length (medians). The warm path is tier load plus tensor restore plus one forward pass.

Tokens	Measured prefill	Predicted	Ratio	Measured warm	Predicted restore
512	111 ms	67 ms	$1.66\times$	101 ms	18 ms
1,024	157 ms	137 ms	$1.15\times$	115 ms	27 ms
2,048	340 ms	288 ms	$1.18\times$	115 ms	44 ms
4,096	878 ms	632 ms	$1.39\times$	146 ms	77 ms
6,144	1,645 ms	1,032 ms	$1.59\times$	181 ms	111 ms

The fixed cost of the measured warm path is the one constant that does *not* transfer: fitting it gives 94.5 ms of overhead plus $13.4\mu\text{s}$ per token, an implied 2.44 GB/s against the 2.0 GB/s the model assumes. The bandwidth transfers; the 94.5 ms does not, because a Colab ephemeral disk read through torch.load is not an NVMe read. N^* must therefore be computed from the storage path actually deployed.

B Admission control

Value Density V2+AC is V2 plus a value-density admission threshold, derived per run rather than hardcoded. It never helped on any workload we ran (Table 3), and the reason is visible in the hit rates: it declines entries at the door without freeing capacity that the eviction policy would not have freed anyway, so it pays cardinality for nothing. Under sustained arrivals the cache sits at near-full utilization regardless.

This is a result about *density*-thresholded admission, and it should not be read as evidence against the N^* -thresholded admission of step (3) in §5, which is a different rule and which we do not evaluate as a standalone policy. It is also conditioned on our accounting flooring sub- N^* hits at zero: a policy that merely declines a worthless hit cannot gain under that floor, and would gain under an accounting that charged the restore.

Table 3: Admission control against plain value density, 10 paired replicates each. Deltas are per-replicate against V2 with 95% t-CIs.

Workload	Policy	Hit%	GPU-s/day	Δ vs V2 [CI]
Synthetic enterprise	ValDensity V2	58.2	73.2	—
	V2+AC	58.2	68.4	-4.8 [-5.3, -4.3]
Azure replay	ValDensity V2	67.0	70.0	—
	V2+AC	66.8	63.9	-6.1 [-6.7, -5.4]

248 C The two evaluation defects

249 Both defects were ours, both flattered a different side, and neither perturbed rankings at random. We
 250 report them because the pattern is the transferable part.

251 *Wall-clock replay (flattered the learned policy)*. An early harness replayed a 6-hour trace at wall-clock
 252 speed and credited hits with unclamped resume-time token counts. Collapsing simulated time drives
 253 session age, idle time, and hour-of-day to approximately zero, which zeroes exactly the features
 254 the predictor was trained on, while the accounting credited a different quantity than the policies
 255 optimized. It showed a learned policy beating LRU. Fixing it reversed the result, and we reported
 256 LRU winning both metrics for some time afterwards.

257 *An architecture-free cost model (flattered LRU)*. The simulator priced prefill with two hardcoded
 258 constants, no model geometry at all, placing the linear/quadratic crossover near 1,300 tokens where
 259 real architectures cross between 12 and 254k. This is not a rescaling, so the usual invariance argument,
 260 that a constant factor cannot change rankings, does not apply: the error inflated an 8k-token session’s
 261 worth relative to a 512-token one by 3–4.5 $\times$, exaggerating precisely the effect the study measures.
 262 Re-pricing from architecture inverted the value comparison, left hit rates untouched, and retracted a
 263 “static value density collapses” finding we had previously reported, which had been a property of the
 264 cost model rather than of the objective.

265 The reason we caught the second at all is that a reader asked why a 598-token prefill was modeled at
 266 57.7 seconds when the hardware measured 37 milliseconds. No test in our suite would have caught it,
 267 because none existed; App. A is the test that does now.

268 D Persona strength

269 Return behavior in our workloads is generated, not observed, and the dispersion σ of per-user return
 270 propensity bounds how much structure any predictor can recover. We sweep it, retraining the predictor
 271 at each setting so the model is as good as that world allows, and re-run the enterprise matrix (Table 4).

Table 4: Persona strength sweep (enterprise, 5 paired seeds, TinyLlama economics). Value is GPU-seconds saved per day, as a ratio to LRU.

σ	Predictor AUC	LRU hit%	V1 $\times$ LRU	V3 $\times$ LRU
0.0	0.659	95.3	0.90	1.34
0.3	0.670	89.5	1.47	1.50
0.6	0.700	80.3	1.66	1.91
1.0	0.739	66.3	1.79	2.25

272 The two claims separate cleanly. The *learned* policy’s standing depends on σ , as it must: at $\sigma=0$,
 273 where users are behaviorally identical, V1 falls to 0.90 $\times$ LRU and the difference is not significant. A
 274 classifier whose premise is per-user prediction should fail when there is no per-user structure, and it
 275 does. The *metric* finding does not depend on σ : V3 beats LRU on value at every setting, and the gap
 276 is smallest exactly where LRU is strongest, which is the regime least favorable to our argument and
 277 still not favorable enough to overturn it.